THE UNIVERSITY OF QUEENSLAND

PROJECT DEVELOPMENT NOTEBOOK

From problematic gaming research to Time Tokens

DECO3500
Design process record
September 2026

Project overview

This notebook records how the project moved from a broad concern about problematic gaming to a tangible prototype that helps people make deliberate choices about work, entertainment and rest. The process is organised by design stage rather than teaching week because several activities overlapped and were revisited.

The project did not treat gaming time as proof of addiction. Literature and interviews repeatedly showed that similar playtime can have different meanings in different lives. The more useful question became whether gaming is chosen deliberately, whether it conflicts with a person's intended priorities, and whether the person can reconsider that choice.

The final design principle is:

Time is not the only variable. Priority judgement is the intervention point.

Four-stage structure

Stage	Focus	Main outcome
1	Domain understanding and interview direction	A genre-agnostic interview plan centred on autonomy, conflict and intended life.
2	Interview organisation and story synthesis	Seven story clusters, then a combined design opportunity from refined Stories 1 and 5.
3	Prototype directions and concept attempts	A comparison of Sea Monster, Deadline Shark, revolver, Conquest Wheel and physical timeline ideas.
4	Final prototype direction and implementation	Time Tokens, a Time Tower concept, and a buildable cuboid flip-to-time prototype.

STAGE 1

Domain Understanding and Interview Direction

From a broad 'gaming addiction' topic to a researchable question about autonomy and priority conflict

1.1 Starting domain

The initial domain was combatting computer game addiction. Our first instinct was to ask why games are so effective at retaining players and how those mechanisms might contribute to excessive or compulsive engagement. The Skinner Box became an early explanatory lens: repeated actions can be strengthened by reinforcement, particularly when rewards are frequent, uncertain or socially meaningful.

This lens was useful for opening the topic, but it was too narrow to explain the full experience described in the literature. Games do not retain people through reward alone. They also create obligations, accumulated investment and relationships that make stopping difficult even when play is no longer fully enjoyable.

1.2 Initial mechanism map

Mechanism	Initial interpretation	What it suggested we should ask
Reinforcement / reward loop	Progress, mastery and variable rewards encourage repeated action.	What makes a player start another session or another match?
Temporal obligation	Daily quests, events and deadlines turn optional play into a scheduled duty.	Does the player feel they will lose something if they do not return?
Sunk cost / collection	Past time, rank, items and completion progress can make leaving feel wasteful.	Does accumulated investment influence the decision to continue?
Social obligation	Teams, guilds and friendships can create belonging and responsibility.	Is continued play a personal choice, a social commitment, or both?

1.3 What gaming can satisfy

The initial reading suggested that gaming can satisfy competence, escape and independence. The same reading also connected problematic play with emotional dependence, social detachment, increasing time, preference for online play, stress, and negative outcomes such as lower academic performance, depression, anxiety and reduced life satisfaction. These findings helped us understand risk, but they did not justify labelling all highly engaged players as addicted.

1.4 Refining the problem definition

We therefore separated four ideas that are often collapsed into one: enjoyment, high engagement, compulsive engagement and disorder. The practical boundary could not be a fixed number of hours. A player might spend many hours on a chosen hobby without major conflict, while another person might play less but repeatedly avoid urgent responsibilities and feel unable to stop.

Question	Why it matters

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Is gaming still an active choice?	Tests autonomy rather than judging the activity itself.
Does gaming conflict with the person's intended life?	Connects play with study, work, sleep, relationships and self-defined goals.
Does the behaviour persist despite harm or regret?	Distinguishes a temporary preference from a difficult-to-change pattern.
Can the person reconsider at meaningful moments?	Points toward an intervention that supports reflection rather than restriction.

1.5 Interview questionnaire: first version

The first questionnaire focused on MOBA players. We expected ranked matches, team dependence and continuous rounds to provide a distinctive account of problematic play. In practice, the questions did not reveal a mechanism unique to MOBA games. Similar experiences could occur in role-playing games, sandbox games, casual games or other online communities. We also could not explain convincingly why MOBA should be the only genre in scope.

1.6 Interview questionnaire: revised direction

We changed to a genre-agnostic, participant-led semi-structured interview. Participants could describe the games they personally cared about, their routines, their reasons for starting and stopping, and the situations in which gaming supported or conflicted with life. The revised interview also included people who played rarely or did not currently play. These negative cases helped us avoid assuming that gaming behaviour has one universal meaning.

- Warm-up: current gaming habits, favourite genres and recent routines.
- Motivation: what the participant seeks from play, including competence, social connection, escape or recovery.
- Decision points: what happens before starting, during continuation and when attempting to stop.
- Conflict: moments when gaming displaced a personally important task or need.
- Counter-examples: periods of high playtime without concern, and low playtime that still felt problematic.
- Design prompts: what kind of support would feel helpful rather than controlling or punitive.

1.7 Research activity

Five team members conducted fourteen in-depth interviews. The sample intentionally included different gaming preferences and different levels of engagement, including participants who did not play games. The aim was not to estimate prevalence. It was to collect contrasting accounts that could challenge the team's early assumptions and reveal useful intervention points.

Stage decision

The project moved away from a genre-specific 'reduce gaming time' brief. The interview direction became: understand how people interpret gaming in relation to their own priorities, and identify moments where they can regain deliberate choice.

Reflection

Starting with the Skinner Box helped us recognise persuasive systems, but it risked reducing players to reward-driven behaviour. Expanding the scope to obligation, identity, context and autonomy made the later interviews more open and less judgemental.

References for Stage 1

Arkes, H. R., & Blumer, C. (1985). The psychology of sunk cost. Organizational Behavior and Human Decision Processes, 35(1), 124–140.

Baumeister, R. F., & Leary, M. R. (1995). The need to belong: Desire for interpersonal attachments as a fundamental human motivation. Psychological Bulletin, 117(3), 497–529.

Frommel, J., & Mandryk, R. L. (2022). Daily quests or daily pests? The benefits and pitfalls of engagement rewards in games. Proceedings of the ACM on Human-Computer Interaction, 6(CHI PLAY), Article 226. https://doi.org/10.1145/3549489

Herrnstein, R. J. (1961). Relative and absolute strength of response as a function of frequency of reinforcement. Journal of the Experimental Analysis of Behavior, 4(3), 267–272.

Iacovides, I., & Mekler, E. D. (2019). The role of gaming during difficult life experiences. Proceedings of CHI 2019.

Kahneman, D., & Tversky, A. (1979). Prospect theory: An analysis of decision under risk. Econometrica, 47(2), 263–291.

Acta Psychologica article used in the team's initial literature note (2023), PII S0001691823002238. https://www.sciencedirect.com/science/article/pii/S0001691823002238

Initial working sources cited in the team note: Klonsky et al. (2016), Krossbakken et al. (2018), and Psychreg (2017). Full bibliographic details should be checked before formal submission.

STAGE 2

Interview Organisation and Story Synthesis

From transcripts to seven stories, then from two refined stories to one design opportunity

2.1 Analysis approach

After transcription, we compared the interviews across participants rather than counting isolated answers. We marked repeated motivations, triggers, stopping difficulties, conflicts and counter-examples. We also preserved contradictions. For example, the same number of gaming hours could be described as meaningful leisure by one participant and avoidance by another.

The seven stories below were early clusters, not final personas and not seven separate design directions. Their purpose was to make the range of experiences visible before the team converged.

2.2 Seven initial story clusters

1. Social gaming changes when people start and stop. A session can continue because leaving affects friends or teammates, not because the individual still wants to play.
2. Competition and immediate feedback support competence. Ranking, improvement and the promise of 'one more game' can make stopping feel premature.
3. Immersion creates an unfinished experience. Players may continue because a world, narrative or task feels incomplete rather than because of an external reward.
4. Collection and self-set goals create commitment. Completing sets, maintaining progress or meeting a personal target can continue after the activity becomes less enjoyable.
5. Live-service systems turn play into routine or obligation. Daily rewards, events and limited-time content can make absence feel like a loss.
6. Gaming can become an attention shelter. It may provide temporary relief from stress, uncertainty or difficult real-world tasks.
7. High playtime is not the same as addiction. The meaning of play depends on autonomy, context, consequences and whether it fits the person's intended life.

2.3 Deepening the stories

The early clusters still described gaming mechanisms more than lived consequences. We therefore returned to the transcripts and developed richer story cards. Each card combined the participant's situation, the apparent behaviour, the hidden tension and a design implication. Two cards became especially useful because they challenged time-based judgement from different directions.

2.4 Refined Story 1

Reduced gaming time does not necessarily mean the gaming problem has been resolved — there can be a separation between behavioural control and the player's relationship with desire.

External limits may reduce visible play without changing why the activity is so compelling. When the restriction disappears, the old pattern may return. A player may also replace gaming with another

avoidance behaviour. This story shifted attention away from suppressing behaviour and toward helping the player understand the choice, the desire and the competing priority.

2.5 Refined Story 5

The same level of gaming engagement can have completely different meanings in different life contexts.

A long session may be a deliberate and satisfying leisure choice for one person. For another, it may be connected to low confidence, stress, avoidance or a lack of rewarding alternatives. Time alone cannot tell us which situation is present. The support must therefore help people interpret their own activity rather than impose one universal threshold.

Figure 1. Refined Stories 1 and 5 converging into a design opportunity. Source: team analysis board.

2.6 Combined insight

Story 1 showed that controlling time does not automatically change the relationship with gaming. Story 5 showed that time cannot reliably diagnose the problem in the first place. Together they led to a clearer intervention point:

Time is not the central variable in anti-addiction design. Priority judgement is.

The design opportunity became helping players compare the immediate reward of continuing a game with the delayed value of study, work, sleep, recovery or other personally meaningful activities. The system should not decide what the user ought to value. It should make the competing choices visible at the moment a decision is still possible.

2.7 Design values carried forward

Value	Meaning for the design
Autonomy	The user chooses the allocation and can change it; the object does not punish or lock access.
Awareness	The design makes current and intended allocation visible before time disappears unnoticed.
Personal priorities	Work, gaming and rest are interpreted relative to the user's own situation.
Reconsideration	The most important interaction is a pause that supports a fresh choice.
Non-judgement	Gaming remains a legitimate activity; the issue is unresolved conflict, not entertainment itself.

Stage decision

The team combined elements from refined Stories 1 and 5. The design would focus on priority conflict and active choice, not on detecting addiction from playtime or enforcing a universal limit.

Reflection

Interview analysis changed the project more than the initial literature map. The negative and contrasting cases made it clear that the system needed to support interpretation. A purely restrictive timer would have reproduced the weakness revealed by Story 1.

Evidence and references for Stage 2

Fourteen de-identified semi-structured interview transcripts collected by five team members.

Team transcription coding, affinity clustering and story-card synthesis recorded in the DECO3500 project workspace.

Refined Story 1 and Story 5 analysis board, reproduced in Figure 1.

STAGE 3

Prototype Directions and Concept Attempts

Testing metaphors for pressure, commitment and time allocation before selecting a clear physical language

3.1 Criteria from the research

The interview insight did not immediately specify a physical form. We used five criteria to compare concepts: the user should set their own commitment; the design should create a visible decision point; progress should be understandable without detailed task measurement; the interaction should be tangible and low-friction; and any social layer should provide awareness rather than competition.

3.2 Sea Monster and Deadline Shark

The first concept family turned deadline pressure into a spatial threat. A boat moved toward a sea monster when no progress was made and toward a lighthouse when the user completed real-world work. A multiplayer version allowed friends to see one another's positions and send lightweight encouragement.

Figure 2. Odyssey Escape: a social sea-monster concept. Concept visualisation produced during ideation; not an engineering drawing.

Figure 3. Deadline Shark physical direction. The boat moves between danger and safety as real-world progress changes.

The metaphor was expressive, but implementation depended on measuring real-world task progress. Hours, subtasks and percentages were difficult to compare across different tasks and people. The threat metaphor could also increase anxiety, while the interface risked becoming a conventional productivity tracker. These problems made the concept less compatible with autonomy and non-judgement.

3.3 Revolver commitment concept

The next concept changed the unit of measurement. Instead of asking the system to estimate objective task progress, the user declared how much effort they were willing to commit on that day.

1. At the beginning of the day, the user considers their current state, deadlines and need for rest, then chooses how much time to invest in real-world tasks.
2. The user loads one to six rounds. Each round represents one work or study unit.
3. During the day, the shark continues to consume the available time.
4. A round can be fired only after the corresponding unit has been completed.
5. At the end of the day, the system checks whether the user completed the commitment they chose for themselves.

This direction was stronger because it measured commitment completion rather than claiming to measure the value of a task. The user chose the amount, so the judgement remained personal and adjustable. The literal weapon metaphor was not retained, but its central mechanism — self-selected units that become tangible — shaped the following concepts.

3.4 Derivative A: Conquest Wheel

The Conquest Wheel translated the loaded rounds into characters placed around a rotating schedule. Each character represented a planned period or activity. When its time arrived, completing the plan allowed the user to launch the character into a central battle. The wheel preserved commitment and progression while replacing the gun with a fantasy-board-game metaphor.

Figure 4. Conquest Wheel (讨伐之轮). Concept visualisation produced during ideation; not an engineering drawing.

The concept offered a strong narrative and clear daily ritual. It also introduced many mechanical parts, character meanings and rules. The spectacle could distract from the reflective choice we wanted to support.

3.5 Derivative B: Time as a Physical Landscape

A second derivative removed the combat metaphor. Modular blocks represented study, gaming, rest and social activity. The user arranged a planned timeline, then built an actual timeline as the day unfolded. The result was a physical comparison between intention and lived experience.

Figure 5. Time as a Physical Landscape. Modular activity blocks compare planned and actual time. Concept visualisation produced during ideation.

This version was simpler and more legible. It treated entertainment and rest as legitimate parts of the day, made planned and actual allocation directly comparable, and reduced punitive language. The physical blocks also preserved the strongest idea from the revolver concept: time commitments could be chosen and handled as discrete units.

3.6 Concept comparison

Direction	What it contributed	Main limitation	Outcome
Sea Monster / Deadline Shark	Visible pressure, movement and social presence	Requires task-progress estimation; may increase anxiety	Not selected
Revolver commitment	Self-selected daily commitment in discrete units	Weapon metaphor and a narrow focus on work completion	Mechanism retained
Conquest Wheel	Planning ritual, time sequence and playful physical action	Mechanically complex and metaphor-heavy	Not selected
Physical Landscape	Clear planned/actual comparison across work, gaming and rest	Large device and too many loose parts	Direction selected

Stage decision

The team selected the Physical Landscape direction as the basis for further development because it communicated time allocation more directly and with less mechanical and narrative complexity. We retained discrete self-chosen units, activity categories, visible accumulation and a possible shared rhythm.

Reflection

The strongest concept was not the most dramatic one. Moving from threat to commitment, then from commitment to visible allocation, aligned the prototype more closely with the interview finding: users need support to judge priorities, not a device that judges them.

References and design materials for Stage 3

Deadline Shark design proposal and team discussion records in the DECO3500 project workspace.

Team-generated concept visualisations: Odyssey Escape, Deadline Shark, Conquest Wheel and Time as a Physical Landscape.

Daily Stack, Copenhagen Institute of Interaction Design tangible interface exploration (2010), recorded by the team as a precedent for physical task and time units.

STAGE 4

Final Prototype Direction and Implementation

Time Tokens and Time Tower, followed by a buildable cuboid flip-to-time interaction

4.1 Final system direction

The Physical Landscape was reduced into a smaller and clearer system: Time Tokens and a Time Tower. Separate tokens represent work or study, entertainment and rest. Each token turns a chosen hour into a tangible unit. The tower collects completed units so the day becomes visible as a mixture of activities rather than a single productivity score.

Figure 6. Final system direction: Time Tokens and Time Tower. Concept visualisation produced during ideation; functional scope is defined separately below.

4.2 Core interaction

Step	User action	System response	Purpose
1. Choose	Select the token that matches the next activity.	The activity icon and available unit are shown.	Make intention explicit.
2. Flip	Turn the token onto its active face.	A one-hour timer begins after stable orientation is detected.	Use a deliberate physical gesture.

Step	User action	System response	Purpose
3. Time	Continue the chosen activity.	The 8×8 display shows elapsed or remaining progress.	Keep time visible without a phone.
4. Notify	Reach the end of the unit.	The display changes and the token vibrates.	Create a natural decision point.
5. Reconsider	Flip back, continue, or choose another token.	The completed unit is recorded for the day.	Restore active choice.
6. Collect	Place or register the completed unit with the tower.	The daily pattern accumulates visually.	Support reflection and optional social awareness.

Flip → Time → Display → Reconsider

4.3 Concept scope and prototype scope

The full system includes several activity tokens, a shared Time Tower and a visual daily summary. The current physical prototype does not need to build every part at once. Its critical research question is whether a tangible flip interaction can make one chosen time unit visible and create a useful moment to reconsider.

Layer	Included now	Can be simulated or deferred
Token	Orientation sensing, timer, 8×8 display, vibration and reset/reselection	Wireless transfer and multiple fully built activity bodies
Tower	Role in the scenario and daily accumulation logic	Physical transfer mechanism, networked multi-user data and final enclosure
Social layer	Awareness rather than ranking	Remote synchronisation and long-term accounts

4.4 Form exploration

Before settling on the final enclosure, we sketched three forms: a capsule with a moving section, a triangular form and a cylinder with simple LEDs. The sketch was useful because it separated the interaction problem from the visual styling problem.

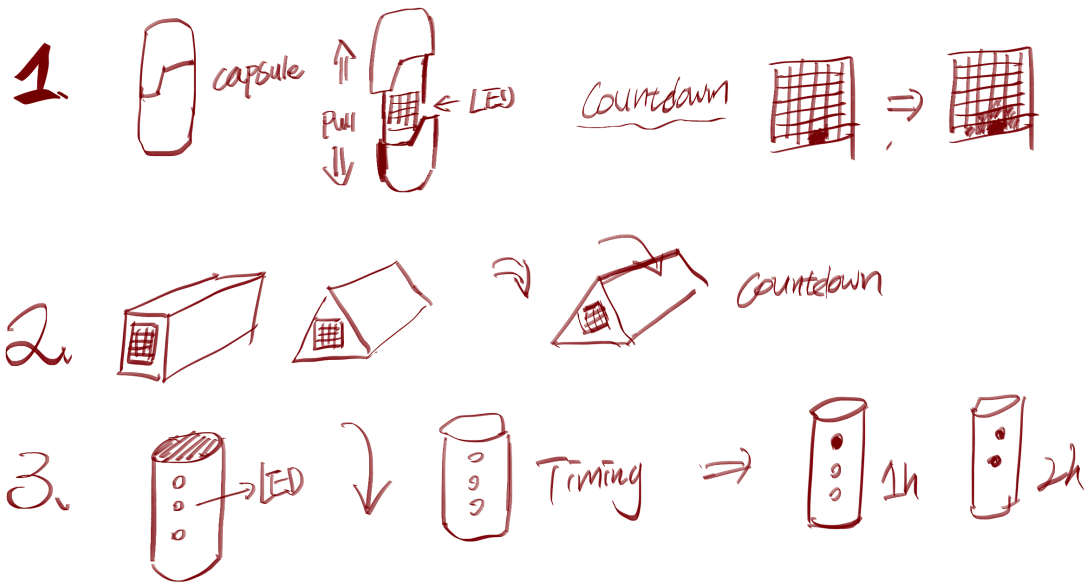

Figure 7. Early form sketches: capsule, triangular body and cylinder.

4.5 Cylinder minimal prototype

The cylinder was first developed as a minimal flip-to-time object. Three LEDs represented accumulated time units and a ball tilt switch provided a simple upright/flipped signal. This version demonstrated the interaction clearly enough to discuss, but it also exposed three limitations: a cylinder can roll, its resting direction is ambiguous, and three individual LEDs can show only a small set of discrete states.

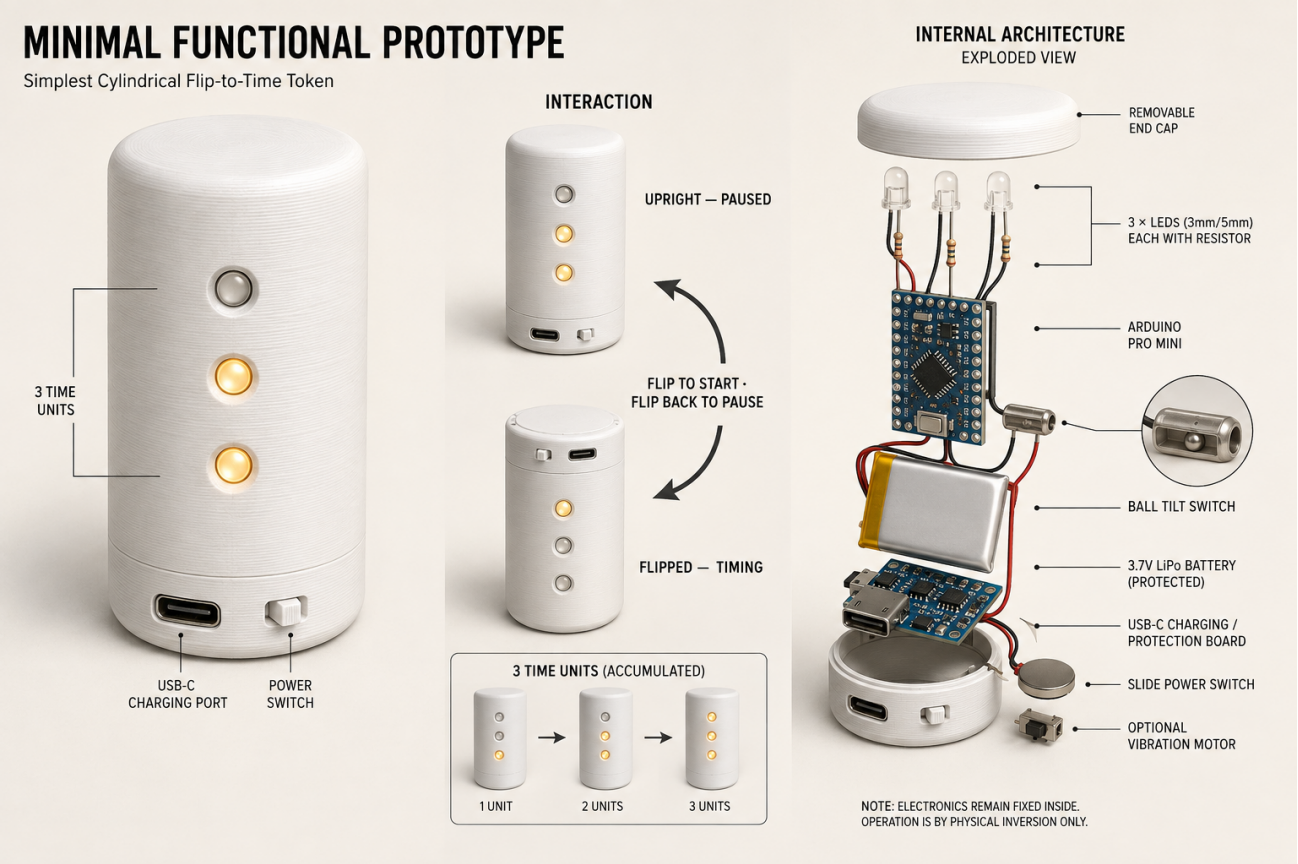

Figure 8. Cylinder-based minimal prototype concept with three LEDs and a ball tilt switch. This was an intermediate design, not the final implementation.

4.6 Capsule twist mechanism exploration

We also examined the Small Battery Box manual as a reference for a capsule that could twist, separate or reveal an internal section. The exploded assembly suggested a compact way to hide electronics and create a satisfying physical transformation.

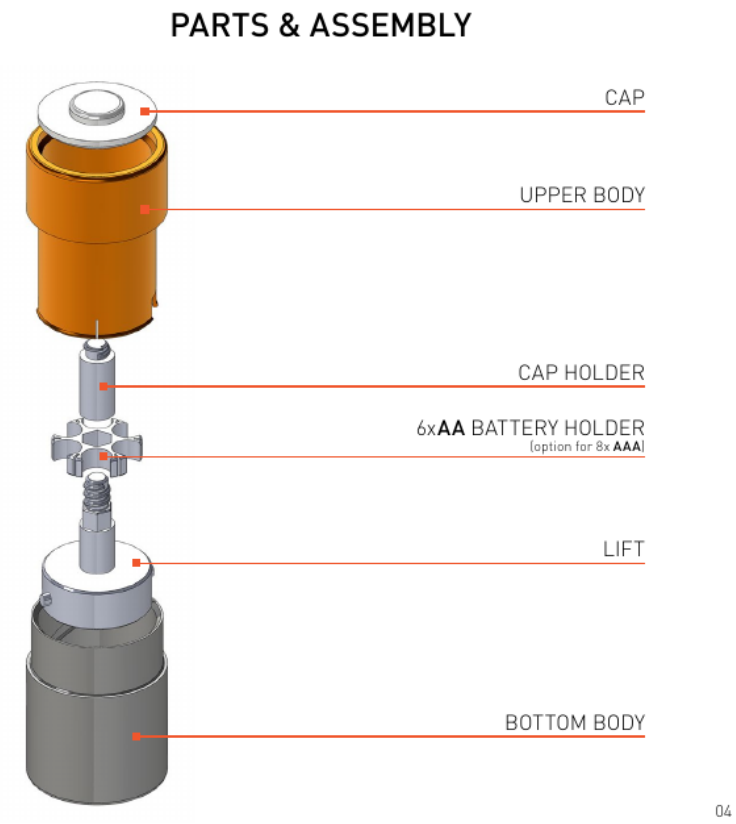

Figure 9. Mechanical reference explored for the capsule direction: Small Battery Box parts and assembly diagram (user manual, p. 4).

The mechanism was not adopted. It required more custom mechanical design, introduced failure points and shifted attention away from the flip interaction that the concept was meant to test. It remains useful evidence of technical exploration, but it is not the final direction.

4.7 Final implementation decisions

Earlier option	Issue found	Final decision
Capsule twist / pull mechanism	Mechanically complex and different from the selected flip interaction	Use a fixed enclosure with no moving outer mechanism
Cylindrical body	Can roll and provides weak orientation cues	Use a rounded cuboid with two stable faces
Three independent LEDs	Only three discrete units and limited feedback vocabulary	Use an 8×8 LED dot matrix for progress, icons and completion states
Ball tilt switch	Only gives a noisy binary tilt signal	Use a tri-axis digital tilt/orientation sensor
Automatic task measurement	Different tasks cannot be compared reliably	Time the commitment chosen by the user
Punitive lock-out	Conflicts with autonomy and does not change desire	Notify and invite reconsideration; do not prevent gaming

4.8 Physical architecture

- Rounded cuboid enclosure with a stable inactive face and stable active face.
- Arduino-compatible microcontroller for timing, orientation logic and display states.
- Tri-axis digital tilt sensor to recognise stable orientation and filter short movements.
- 8×8 LED dot matrix to show the activity icon, running progress, completion and reset states. Activity meaning should not rely on colour alone.
- Small vibration motor for end-of-unit feedback.
- Rechargeable or removable power source, power switch and accessible charging or programming connection.

4.9 Interaction state logic

State	Entry condition	Behaviour	Exit condition
Ready	Token is upright and stable	Display shows activity identity and availability	Stable flip to active face
Running	Active orientation confirmed	Timer runs; matrix updates progress	Unit completes or user flips back
Complete	Chosen unit reaches zero	Completion pattern and vibration	User acknowledges by flipping
Paused / cancelled	User flips back before completion	Elapsed state is held or cancelled according to test setting	Flip again or reset
Review	Completed unit recorded	Daily count is shown locally or transferred to the tower simulation	Select the next activity

4.10 Prototype test focus

1. Do users understand that flipping begins a chosen time commitment without being told repeatedly?
2. Does the 8×8 progress display remain readable at a glance and without colour dependence?
3. Does the end-of-unit vibration create a useful reconsideration moment rather than an interruption?
4. Can users change activity or rest without feeling that the device is judging them?
5. Does the visible accumulation help users discuss planned and actual priorities?
6. Is the cuboid stable, comfortable to handle and reliably recognised by the tri-axis sensor?

Stage decision

The final build direction is a rounded cuboid Time Token using an 8×8 LED dot matrix and a tri-axis digital tilt sensor. The essential prototype cycle is Flip → Time → Display → Reconsider. The Time Tower remains the larger system concept and can be represented through a partial physical model or a controlled simulation during early evaluation.

Reflection

Technical simplification improved conceptual fidelity. Removing the twist mechanism, rolling cylinder and three-LED limit did not reduce the research idea; it made the autonomy and reconsideration mechanism

easier to test. The next prototype should be judged by whether it supports a meaningful choice, not by whether every visual feature in the concept rendering is already functional.

References and technical materials for Stage 4

Small Battery Box — User Manual, v1. User-supplied mechanical reference, 17 pages.

Team form sketches and cylinder prototype visualisation reproduced in Figures 7 and 8.

Time Tokens and Time Tower concept visualisation reproduced in Figure 6.

Arduino-compatible prototyping platform, 8×8 LED dot matrix module, tri-axis digital tilt/orientation sensor and vibration motor identified during implementation planning. Exact component datasheets should be added once the final purchased modules are confirmed.

Decision trail at a glance

Move	Reason
Gaming addiction → problematic engagement	Avoid assuming that hours alone define the problem.
MOBA-only interview → genre-agnostic interview	MOBA-specificity was not supported by the first questionnaire.
Seven clusters → refined Stories 1 and 5	The two stories exposed the limits of time control and time-based judgement.
Reduce playtime → support priority judgement	The design should help users compare immediate and delayed value.
Deadline Shark → self-selected commitment	Objective task progress was difficult to measure and compare.
Revolver → Physical Landscape	Keep discrete commitments while removing the weapon metaphor.
Physical Landscape → Time Tokens	Reduce size and complexity while preserving tangible allocation.
Cylinder / twist → cuboid flip token	Improve stability, orientation sensing and conceptual focus.
Three LEDs / ball switch → 8×8 matrix / tri-axis sensor	Expand feedback and make orientation detection more reliable.

This trail is the main argument of the notebook: each prototype decision responds to a finding, a contradiction or an implementation test. The final object is therefore not simply a timer. It is a tangible prompt for deliberate allocation and repeated reconsideration.